MATH 327 Homework 1 Fall 2026

Yazan Ismail

10/7/2026

1. Problem 1.3

Problem. Prove that $1^3 + 2^3 + \cdots + n^3 = (1 + 2 + \cdots + n)^2$ for all $n \in \mathbb{N}$.

Lemma 1. *For all $n \in \mathbb{N}$,*

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

Proof. Let $P(n)$ be the statement

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

We proceed by induction on n.

Base case. For $n = 1$,

$$1 = \frac{1(1+1)}{2},$$

so $P(1)$ holds.

Inductive step. Suppose $P(m)$ holds for some $m \in \mathbb{N}$. Thus,

$$1 + 2 + \cdots + m = \frac{m(m+1)}{2}. \tag{1}$$

Adding $m + 1$ to both sides of (1), we get

$$\begin{aligned} 1 + 2 + \cdots + m + (m+1) &= \frac{m(m+1)}{2} + (m+1) \\ &= (m+1)\left(\frac{m}{2} + 1\right) \\ &= (m+1)\left(\frac{m+2}{2}\right) \\ &= \frac{(m+1)(m+2)}{2}. \end{aligned}$$

Therefore, $P(m+1)$ holds, and the result follows by induction. □

Theorem 1. *For all* $n \in \mathbb{N}$,

$$1^3 + 2^3 + \cdots + n^3 = (1 + 2 + \cdots + n)^2.$$

Proof. Let $P(n)$ be the statement

$$1^3 + 2^3 + \cdots + n^3 = (1 + 2 + \cdots + n)^2.$$

We proceed by induction on n.

Base case. For $n = 1$,

$$1^3 = 1^2,$$

so $P(1)$ holds.

Inductive step. Suppose $P(m)$ holds for some $m \in \mathbb{N}$. Thus,

$$1^3 + 2^3 + \cdots + m^3 = (1 + 2 + \cdots + m)^2. \tag{2}$$

Adding $(m+1)^3$ to both sides of (2), we get

$$\begin{aligned}
1^3 + 2^3 + \cdots + m^3 + (m+1)^3 &= (1 + 2 + \cdots + m)^2 + (m+1)^3 \\
&= \left(\frac{m(m+1)}{2}\right)^2 + (m+1)^3 && \text{by Lemma 1} \\
&= (m+1)^2 \left(\frac{m^2}{4} + m + 1\right) \\
&= (m+1)^2 \left(\frac{m^2 + 4m + 4}{4}\right) \\
&= (m+1)^2 \left(\frac{m+2}{2}\right)^2 \\
&= \left(\frac{(m+1)(m+2)}{2}\right)^2 \\
&= (1 + 2 + \cdots + m + (m+1))^2 && \text{by Lemma 1.}
\end{aligned}$$

Therefore, $P(m+1)$ holds, and the result follows by induction. □

2. Problem 1.12(c)

Problem. Use induction and part (b), $\binom{n}{k} + \binom{n}{k-1} = \binom{n+1}{k}$ for $k = 1, 2, \ldots, n$, to prove the binomial theorem: for all $a, b \in \mathbb{R}$ and $n \in \mathbb{N}$,

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k.$$

The following supporting identity is part 1.12(b).

Lemma 2 (Pascal identity from 1.12(b))**.** *For $n \in \mathbb{N}$ and $k = 1, 2, \ldots, n$,*

$$\binom{n}{k} + \binom{n}{k-1} = \binom{n+1}{k}.$$

Proof. Using the definition of the binomial coefficients,

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}, \qquad \binom{n}{k-1} = \frac{n!}{(k-1)!(n-k+1)!},$$

and

$$\binom{n+1}{k} = \frac{(n+1)!}{k!(n-k+1)!}.$$

Therefore,

$$\begin{aligned}
\binom{n}{k} + \binom{n}{k-1} &= \frac{n!}{k!(n-k)!} + \frac{n!}{(k-1)!(n-k+1)!} \\
&= n! \left(\frac{n-k+1}{k!(n-k+1)!} + \frac{k}{k!(n-k+1)!} \right) \\
&= n! \left(\frac{n+1}{k!(n-k+1)!} \right) \\
&= \frac{(n+1)!}{k!(n-k+1)!} \\
&= \binom{n+1}{k}.
\end{aligned}$$

□

Theorem 2 (Binomial theorem)**.** *For all $a, b \in \mathbb{R}$ and $n \in \mathbb{N}$,*

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k.$$

Proof. Let $P(n)$ be the stated binomial identity. We proceed by induction on n.

Base case. Let $n = 1$. Then $P(1)$ states

$$\begin{aligned}(a+b)^1 &= \binom{1}{0}a + \binom{1}{1}b \\ &= a + b.\end{aligned}$$

Inductive step. Suppose $P(m)$ holds for some $m \in \mathbb{N}$. Thus, we have

$$\begin{aligned}(a+b)^m &= \binom{m}{0}a^m + \binom{m}{1}a^{m-1}b \\ &\quad + \binom{m}{2}a^{m-2}b^2 + \cdots \\ &\quad + \binom{m}{m-1}ab^{m-1} + \binom{m}{m}b^m \\ &= a^m + ma^{m-1}b + \frac{1}{2}m(m-1)a^{m-2}b^2 \\ &\quad + \cdots + mab^{m-1} + b^m.\end{aligned}$$

In these expanded expressions, only terms with valid indices are present when m is small.

$$
\begin{aligned}
(a+b)^{m+1} &= (a+b)^m(a+b) \\
&= (a+b)\Big[\binom{m}{0}a^m + \binom{m}{1}a^{m-1}b \\
&\qquad + \binom{m}{2}a^{m-2}b^2 + \cdots \\
&\qquad + \binom{m}{m-1}ab^{m-1} + \binom{m}{m}b^m\Big] \\
&= \binom{m}{0}a^{m+1} + \binom{m}{1}a^m b \\
&\quad + \binom{m}{2}a^{m-1}b^2 + \cdots \\
&\quad + \binom{m}{m-1}a^2 b^{m-1} + \binom{m}{m}ab^m \\
&\quad + \binom{m}{0}a^m b + \binom{m}{1}a^{m-1}b^2 \\
&\quad + \binom{m}{2}a^{m-2}b^3 + \cdots \\
&\quad + \binom{m}{m-1}ab^m + \binom{m}{m}b^{m+1} \\
&= \binom{m}{0}a^{m+1} \\
&\quad + \left(\binom{m}{1} + \binom{m}{0}\right)a^m b \\
&\quad + \left(\binom{m}{2} + \binom{m}{1}\right)a^{m-1}b^2 \\
&\quad + \cdots \\
&\quad + \left(\binom{m}{m} + \binom{m}{m-1}\right)ab^m \\
&\quad + \binom{m}{m}b^{m+1}.
\end{aligned}
$$

By part (b),

$$
\binom{m}{k} + \binom{m}{k-1} = \binom{m+1}{k}.
$$

Also,

$$\binom{m}{0} = \frac{m!}{0!(m-0)!} = 1 = \binom{m+1}{0},$$

$$\binom{m}{m} = \frac{m!}{m!(m-m)!} = 1 = \binom{m+1}{m+1}.$$

Hence,

$$\begin{aligned}(a+b)^{m+1} &= \binom{m+1}{0}a^{m+1} \\ &\quad + \binom{m+1}{1}a^m b \\ &\quad + \binom{m+1}{2}a^{m-1}b^2 + \cdots \\ &\quad + \binom{m+1}{m}ab^m \\ &\quad + \binom{m+1}{m+1}b^{m+1} \\ &= a^{m+1} + (m+1)a^m b \\ &\quad + \frac{1}{2}(m+1)ma^{m-1}b^2 \\ &\quad + \cdots + (m+1)ab^m + b^{m+1}.\end{aligned}$$

Thus, $P(m)$ implies $P(m+1)$. By induction, $P(n)$ is true for all $n \in \mathbb{N}$, so the binomial theorem holds. □

3. Bernoulli's inequality

Problem. Use induction to show that for any $x \geq -1$ and any $n \in \mathbb{N}$ we have

$$(1+x)^n \geq 1 + nx.$$

(Why is there a restriction on x? Did you use it in your proof?)

Theorem 3. *For every $x \geq -1$ and every $n \in \mathbb{N}$,*

$$(1+x)^n \geq 1 + nx.$$

Proof. Fix $x \in [-1, \infty)$, and let $P(n)$ be the statement

$$(1+x)^n \geq 1 + nx.$$

We prove $P(n)$ for all $n \in \mathbb{N}$ by induction.

Base case. For $n = 1$,

$$(1+x)^1 = 1 + x = 1 + 1x,$$

so (P(1)) holds.

Inductive step. Suppose $P(m)$ holds for some $m \in \mathbb{N}$. Thus,

$$(1+x)^m \geq 1 + mx.$$

Since $x \geq -1$, we have $1 + x \geq 0$. Multiplying both sides by (1+x) preserves the inequality:

$$\begin{aligned}(1+x)^{m+1} &\geq (1+mx)(1+x) \\ &= 1 + (m+1)x + mx^2.\end{aligned}$$

Because $m \in \mathbb{N}$ and $x^2 \geq 0$, we have $mx^2 \geq 0$. Hence,

$$(1+x)^{m+1} \geq 1 + (m+1)x.$$

Thus (P(m+1)) holds, and the result follows by induction. □

4. Problem 2.4

Problem. Show that $\sqrt[3]{5-\sqrt{3}}$ is not a rational number.

Lemma 3. *$\sqrt{3}$ is irrational.*

Proof. Suppose, for the sake of contradiction, that $\sqrt{3}$ is rational. Then

$$\sqrt{3} = \frac{p}{q}$$

for some $p \in \mathbb{Z}$ and $q \in \mathbb{N}$. We may assume that p and q are relatively prime. Squaring both sides gives

$$3q^2 = p^2.$$

We consider the possible parities of p and q.

Case 1: p and q are both odd. Write $p = 2k+1$ and $q = 2j+1$ for some $k, j \in \mathbb{Z}$. Then

$$3(2j+1)^2 = (2k+1)^2.$$

Expanding and simplifying,

$$12j^2 + 12j + 3 = 4k^2 + 4k + 1,$$
$$6j^2 + 6j + 1 = 2k^2 + 2k.$$

The left side of the last equality is odd, while the right side is even, a contradiction.

Case 2: p and q have opposite parity. Then p^2 and q^2 have opposite parity. Since $3q^2$ has the same parity as q^2, it cannot equal p^2, a contradiction.

Case 3: p and q are both even. Then p and q have a common factor of 2, contradicting the assumption that they are relatively prime.

These cases exhaust all possible parities of p and q. Therefore, $\sqrt{3}$ is irrational. □

Theorem 4. *$\sqrt[3]{5-\sqrt{3}}$ is irrational.*

Proof. Suppose, for the sake of contradiction, that $\sqrt[3]{5-\sqrt{3}}$ is rational. Then

$$\sqrt[3]{5-\sqrt{3}} = \frac{p}{q}$$

for some $p \in \mathbb{Z}$ and $q \in \mathbb{N}$. Cubing both sides gives

$$5 - \sqrt{3} = \frac{p^3}{q^3}.$$

Rearranging,

$$\sqrt{3} = \frac{5q^3 - p^3}{q^3}.$$

Since $5q^3 - p^3 \in \mathbb{Z}$ and $q^3 \in \mathbb{N}$, this would make $\sqrt{3}$ rational, contradicting Lemma 3. Therefore, $\sqrt[3]{5-\sqrt{3}}$ is irrational. □

5. Problem 3.6(b)

Problem. Use induction to prove that $|a_1+a_2+\cdots+a_n| \leq |a_1|+|a_2|+\cdots+|a_n|$ for any n real numbers $a_1, a_2, \ldots, a_n$.

Theorem 5. *For every $n \in \mathbb{N}$ and all $a_1, a_2, \ldots, a_n \in \mathbb{R}$,*

$$|a_1 + a_2 + \cdots + a_n| \leq |a_1| + |a_2| + \cdots + |a_n|.$$

Proof. Let $P(n)$ be the statement that, for every $a_1, a_2, \ldots, a_n \in \mathbb{R}$,

$$|a_1 + a_2 + \cdots + a_n| \leq |a_1| + |a_2| + \cdots + |a_n|.$$

We proceed by induction on n.

Base case. For $n = 1$ and any $a_1 \in \mathbb{R}$,

$$|a_1| \leq |a_1|.$$

Thus $P(1)$ holds.

Inductive step. Suppose $P(m)$ holds for some $m \in \mathbb{N}$. Let $a_1, a_2, \ldots, a_{m+1} \in \mathbb{R}$. By the induction hypothesis,

$$|a_1 + a_2 + \cdots + a_m| \leq |a_1| + |a_2| + \cdots + |a_m|. \tag{3}$$

Adding $|a_{m+1}|$ to both sides of (3) gives

$$|a_1 + a_2 + \cdots + a_m| + |a_{m+1}| \leq |a_1| + |a_2| + \cdots + |a_m| + |a_{m+1}|.$$

By the triangle inequality,

$$|(a_1 + a_2 + \cdots + a_m) + a_{m+1}| \leq |a_1 + a_2 + \cdots + a_m| + |a_{m+1}|.$$

Therefore, by transitivity,

$$|a_1 + a_2 + \cdots + a_m + a_{m+1}| \leq |a_1| + |a_2| + \cdots + |a_m| + |a_{m+1}|.$$

Thus $P(m+1)$ holds, and the result follows by induction. □

6. Bounds table for Question 4.1

Problem. Fill in the upper bounds, lower bounds, maximum, minimum, supremum, and infimum for the sets (h), (i), (j), (k), (n), (r), and (w) from Question 4.1.

The upper- and lower-bound entries below give all real bounds for each set.

Set	Upper bounds	Lower bounds	Maximum	Minimum	Supremum	Infimum
(h)	DNE	$\{x \in \mathbb{R} : x \leq 2\}$	DNE	2	DNE	2
(i)	$\{x \in \mathbb{R} : x \geq 1\}$	$\{x \in \mathbb{R} : x \leq 0\}$	1	0	1	0
(j)	$\{x \in \mathbb{R} : x \geq 1\}$	$\{x \in \mathbb{R} : x \leq \frac{2}{3}\}$	DNE	$\frac{2}{3}$	1	$\frac{2}{3}$
(k)	DNE	$\{x \in \mathbb{R} : x \leq 0\}$	DNE	0	DNE	0
(n)	$\{x \in \mathbb{R} : x \geq \sqrt{2}\}$	$\{x \in \mathbb{R} : x \leq -\sqrt{2}\}$	DNE	DNE	$\sqrt{2}$	$-\sqrt{2}$
(r)	$\{x \in \mathbb{R} : x \geq 1\}$	$\{x \in \mathbb{R} : x \leq 1\}$	1	1	1	1
(w)	$\{x \in \mathbb{R} : x \geq \frac{\sqrt{3}}{2}\}$	$\{x \in \mathbb{R} : x \leq -\frac{\sqrt{3}}{2}\}$	$\frac{\sqrt{3}}{2}$	$-\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{2}$	$-\frac{\sqrt{3}}{2}$

7. Problem 4.7

Problem. Let S and T be nonempty bounded subsets of $\mathbb{R}$.

(a) Prove that if $S \subseteq T$, then $\inf T \leq \inf S \leq \sup S \leq \sup T$.

(b) Prove that $\sup(S \cup T) = \max\{\sup S, \sup T\}$. (Find a way to use (a) to prove (b).)

Theorem 6 (Part (a))**.** *Let S and T be nonempty bounded subsets of $\mathbb{R}$. If $S \subseteq T$, then*

$$\inf T \leq \inf S \leq \sup S \leq \sup T.$$

Proof. Suppose $S \subseteq T$.
Supremum case. Suppose, for the sake of contradiction, that $\sup S > \sup T$. Since $\sup T$ is an upper bound for T, every $s \in S$ also satisfies $s \leq \sup T$. Thus $\sup T$ is an upper bound for S. But $\sup T < \sup S$, contradicting the fact that $\sup S$ is the least upper bound for S. Hence $\sup S \leq \sup T$.
Infimum case. Suppose, for the sake of contradiction, that $\inf S < \inf T$. Since $\inf T$ is a lower bound for T, every $s \in S$ also satisfies $\inf T \leq s$. Thus $\inf T$ is a lower bound for S. But $\inf T > \inf S$, contradicting the fact that $\inf S$ is the greatest lower bound for S. Hence $\inf T \leq \inf S$.

Since S is nonempty, choose $s \in S$. The definitions of infimum and supremum give $\inf S \leq s \leq \sup S$, so $\inf S \leq \sup S$. Combining the three inequalities yields

$$\inf T \leq \inf S \leq \sup S \leq \sup T.$$

□

Theorem 7 (Part (b))**.** *Let S and T be nonempty bounded subsets of $\mathbb{R}$. Then*

$$\sup(S \cup T) = \max\{\sup S, \sup T\}.$$

Proof. Let $x = \max\{\sup S, \sup T\}$ and $\alpha = \sup(S \cup T)$. Since $S \subseteq S \cup T$ and $T \subseteq S \cup T$, part (a) gives

$$\sup S \leq \alpha \qquad \text{and} \qquad \sup T \leq \alpha.$$

Therefore $x \leq \alpha$.

Suppose, for the sake of contradiction, that $x < \alpha$. Then $0 < \alpha - x$, so

$$0 < \frac{\alpha - x}{2} < \alpha - x.$$

Adding x gives

$$x < x + \frac{\alpha - x}{2} < \alpha.$$

Since $x \geq \sup S$ and $x \geq \sup T$, x is an upper bound for both S and T. Consequently, $x + \frac{\alpha - x}{2}$ is an upper bound for $S \cup T$. This contradicts the fact that α is the least upper bound for $S \cup T$. Thus $x = \alpha$, or

$$\sup(S \cup T) = \max\{\sup S, \sup T\}.$$

□

8. Problem 4.8(b)

Problem. Let S and T be nonempty subsets of $\mathbb{R}$ such that $s \leq t$ for all $s \in S$ and $t \in T$. Prove that $\sup S \leq \inf T$.

Theorem 8. *Let S and T be nonempty subsets of $\mathbb{R}$ such that*

$$s \leq t \qquad \text{for all } s \in S \text{ and } t \in T.$$

Then $\sup S \leq \inf T$.

Proof. Let S and T be nonempty subsets of $\mathbb{R}$ with the following property:

$$s \leq t \quad \text{for all } s \in S \text{ and } t \in T.$$

Suppose for the sake of contradiction that $\sup(S) > \inf(T)$. Thus, $\inf(T)$ is not an upper bound for S. Hence, there exists an $s \in S$ such that $\inf(T) < s < \sup(S)$. Let s_1 be such s. Now we have two cases:

Case 1. s_1 is a lower bound for T, in which case, s_1 is a greater lower bound for T than $\inf(T)$, which is a contradiction.

Case 2. $s_1 \in T$. Since $s_1 \leq t$ for all $s \in S$ and $t \in T$, then $\min(T) = s_1$. Since s_1 is the minimum of T, then $s_1 = \inf(T)$, contradicting $\inf(T) < s_1$.

Case 3. s_1 is an upper bound for T, in which case, for all $t \in T$, $t \leq s_1$. If $t < s_1$ for any $t \in T$, then we contradict $s \leq t$ for all $s \in S$ and $t \in T$. Thus, $t = s_1$ for all $t \in T$. However, since $s_1 < \sup(S)$, then there exists $s_2 \in S$ such that $s_1 < s_2 < \sup(S)$. Thus, $s_2 \in S$ such that $s_2 > t$ for all $t \in T$, contradicting our supposition that for all $s \in S$ and for all $t \in T$, $s \leq t$.

Hence, we've exhausted all cases with contradictions.

Therefore, $\sup(S) \leq \inf(T)$. □